

In Exercises 49 through 54, find the value of the function where $a = 6$ and $b = 4$.

49. `int(-a / 2)`

50. `round(a / b)`

51. `abs(a - 5)`

52. `abs(4 - a)`

53. `round(a + 0.5)`

54. `int(b * 0.5)`

In Exercises 55 through 60, rewrite the statements using augmented assignment operators.

55. `cost = cost + 5`

56. `sum = sum * 2`

57. `cost = cost / 6`

58. `sum = sum - 7`

59. `sum = sum % 2`

60. `cost = cost // 3`

In Exercises 61 through 68, write a program that has one line of code for each step.

61. Calculate Profit The following steps calculate a company's profit.

- (a) Create the variable *revenue* and assign it the value 98,456.
- (b) Create the variable *costs* and assign it the value 45,000.
- (c) Create the variable *profit* and assign it the difference between the values of the variables *revenue* and *costs*.
- (d) Display the value of the variable *profit*.

62. Stock Purchase The following steps calculate the amount of a stock purchase.

- (a) Create the variable *costPerShare* and assign it the value 25.625.
- (b) Create the variable *numberOfShares* and assign it the value 400.
- (c) Create the variable *amount* and assign it the product of the values of *costPerShare* and *numberOfShares*.
- (d) Display the value of the variable *amount*.

63. Discounted Price The following steps calculate the price of an item after a 30% reduction.

- (a) Create the variable *price* and assign it the value 19.95.
- (b) Create the variable *discountPercent* and assign it the value 30.
- (c) Create the variable *markdown* and assign it the value of (*discountPercent* divided by 100) times the value of *price*.
- (d) Decrease the value of *price* by *markdown*.
- (e) Display the value of *price* (rounded to two decimal places).

64. Break-Even Point The following steps calculate a company's break-even point, the number of units of goods the company must manufacture and sell in order to break even.

- (a) Create the variable *fixedCosts* and assign it the value 5,000.
- (b) Create the variable *pricePerUnit* and assign it the value 8.
- (c) Create the variable *costPerUnit* and assign it the value 6.
- (d) Create the variable *breakEvenPoint* and assign it the value of *fixedCosts* divided by (the difference of the values of *pricePerUnit* and *costPerUnit*).
- (e) Display the value of the variable *breakEvenPoint*.

65. Savings Account The following steps calculate the balance after three years when \$100 is deposited in a savings account at 5% interest compounded annually.

- (a) Create the variable *balance* and assign it the value 100.
- (b) Increase the value of the variable *balance* by 5%.
- (c) Increase the value of the variable *balance* by 5%.

- (d) Increase the value of the variable *balance* by 5%.
 - (e) Display the value of *balance* (rounded to two decimal places).
- 66. Savings Account** The following steps calculate the balance at the end of three years when \$100 is deposited at the beginning of each year in a savings account at 5% interest compounded annually.
- (a) Create the variable *balance* and assign it the value 100.
 - (b) Increase the value of the variable *balance* by 5%, and add 100 to it.
 - (c) Increase the value of the variable *balance* by 5%, and add 100 to it.
 - (d) Increase the value of the variable *balance* by 5%.
 - (e) Display the value of *balance* (rounded to two decimal places).
- 67. Savings Account** The following steps calculate the balance after 10 years when \$100 is deposited in a savings account at 5% interest compounded annually.
- (a) Create the variable *balance* and assign it the value 100.
 - (b) Multiply the value of the variable *balance* by 1.05 raised to the 10th power.
 - (c) Display the value of *balance* (rounded to two decimal places).
- 68. Profit from Stock** The following steps calculate the percentage profit from the sale of a stock.
- (a) Create the variable *purchasePrice* and assign it the value 10.
 - (b) Create the variable *sellingPrice* and assign it the value 15.
 - (c) Create the variable *percentProfit* and assign it 100 times the value of the difference between *sellingPrice* and *purchasePrice* divided by *purchasePrice*.
 - (d) Display the value of the variable *percentProfit*.

In Exercises 69 through 78, write a program to solve the problem and display the answer. The program should use variables for each of the quantities.

- 69. Corn Production** Suppose each acre of farmland produces 18 tons of corn. How many tons of corn can be grown on a 30-acre farm?
- 70. Projectile Motion** Suppose a ball is thrown straight up in the air with an initial velocity of 50 feet per second and an initial height of 5 feet. How high will the ball be after 3 seconds? **Note:** The height after t seconds is given by the expression $-16t^2 + v_0t + h_0$, where v_0 is the initial velocity and h_0 is the initial height.
- 71. Distance Covered** If a car left the airport at 5 o'clock and arrived home at 9 o'clock, what was the distance covered? **Note:** Speed of the car is 81.34 km per hour.
- 72. Gas Mileage** A motorist wants to determine her gas mileage. At 23,352 miles (on the odometer) the tank is filled. At 23,695 miles the tank is filled again with 14 gallons. How many miles per gallon did the car average between the two fillings?
- 73. Power Usage** A survey showed that the average monthly electricity consumption for a city was 750 million watts per month. What was the daily power consumption in watts of each resident? **Note:** The city has a population of about 5 million people.
- 74. Square Deck** José is building a square deck at the back of his house. José has a building permit for a 432-square-foot deck. How long will each side of the deck be?
- 75. Banks** A bank offers 8.7% interest per year on all savings accounts. If a savings account initially contains \$1000, how much money will the account hold two years later?

- 76. Population Increase** You grew up in a tiny village and had to move to a nearby city for your undergraduate. When you left, the population was 845. You recently heard that the population of your village has grown by 6.5%. What is the present population of the village? Round the population to the nearest whole number.
- 77. Bacterial Growth** Suppose a surface initially contained $2.19 \cdot 10^{14}$ bacterial cells. After some time, the surface contained $4.68 \cdot 10^{14}$ bacterial cells. Calculate the percentage of bacterial growth. Display the answer rounded to the nearest whole number.
- 78. Calories** Estimate the number of calories in one cubic mile of chocolate ice cream. **Note:** There are 5,280 feet in a mile and one cubic foot of chocolate ice cream contains about 48,600 calories.

Solutions to Practice Problems 2.1

6. Modulus operations are performed before subtractions. If the intent is for the subtraction to be performed first, the expression should be written $(7 - 4) \% 3$.
- The first assignment statement assigns the value of the variable `var2` to the variable `var1`, whereas the second assignment statement assigns `var1`'s value to `var2`.
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	a	b	c
a = 5	5	does not exist	does not exist
b = 4	5	4	does not exist
c = a * b	5	4	20
a = c // a	4	4	20
print((a - b)*c)	4	4	20
b = b * b * b	4	64	20

Each time an assignment statement is executed, only one variable (the variable to the left of the equal sign) has its value changed.

- Each of the following four statements increases the value of `var` by 5%.

```
var = var + (.05 * var)
var = 1.05 * var
var += .05 * var
var *= 1.05
```